

Lab Report 01 — The Grad–Shafranov Residual Scale

A field-strength bias in the loss and the KPI, and what removing it bought

2026-07-28

1. Hypothesis

Every configuration’s residual is divided by a per-configuration constant before it reaches the loss or the KPI:

$$R_{GS} = \frac{\Delta^* \psi + \mu_0 R^2 p'(\psi) + F(\psi) F'(\psi)}{\text{scale}}, \quad \text{scale} = \left(\frac{F_{axis}}{R_0} \right)^2 + 1$$

The divisor is dimensionally mismatched to the term it normalises, biasing both measurement and training by F_{axis} . If so, the KPI ranks configurations by field strength rather than solution quality — and since the objective is median + 0.3 p_{95} , the tail fills with configurations that are not actually worse. The loss inherits the same weighting.

Context: a 5×256 hard-BC MLP had plateaued at $\sim 2.5 \times 10^{-4}$ median $|R_{GS}|$, unmoved by width, depth, or epochs.

2. Test

Spearman rank correlation against F_{axis} of three quantities, on two frozen checkpoints (5×256 , 266k params; 5×200 , 163k params) over the frozen protocol — 200 Sobol configurations $\times$ 8,192 area-uniform points:

1. the reported KPI, median $|R_{GS}|$;
2. the dominant right-hand-side term it balances, $|FF'|/\text{scale}$;
3. the scale-free ratio $|R_{GS}|/|FF'|$.

If the KPI’s F_{axis} trend is real model behaviour, it survives dividing the balanced term out. If it is the divisor, it disappears.

3. Findings

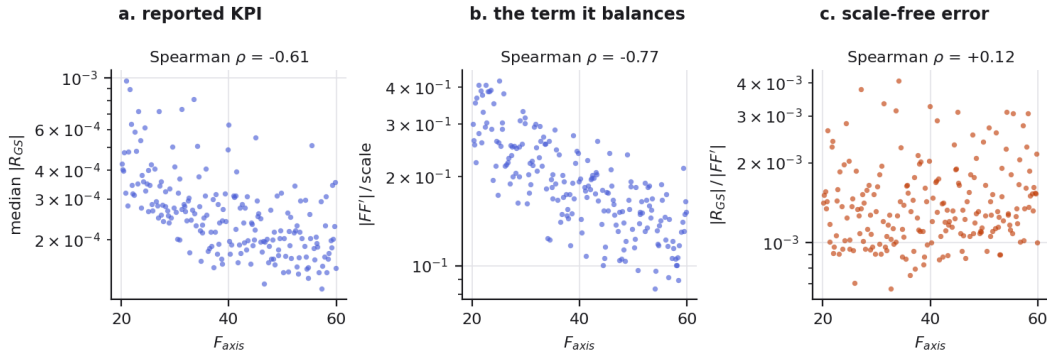


Figure 1: The KPI’s trend with F_{axis} survives in the term being balanced, not in the scale-free error.

vs. F_{axis}	5×256	5×200
reported median $ R_{GS} $	$\rho = -0.61$	$\rho = -0.51$
normalised field term $ FF' /\text{scale}$	$\rho = -0.77$	$\rho = -0.77$
scale-free error $ R_{GS} / FF' $	$\rho = +0.12$	$\rho = +0.17$

The divisor correlates -0.77 with the term it normalises. Dividing that term out removes the trend. Hypothesis confirmed.

Two limits on the claim. The scale-free error still spans $6.1 \times / 7.3 \times$ against $7.9 \times / 9.3 \times$ raw — the divisor explains the *trend*, not the whole spread. The remaining scatter tracks shape (elongation $\rho = +0.45$, triangularity $\rho = +0.30$), not profile parameters ($|\rho| \leq 0.08$ for p_0).

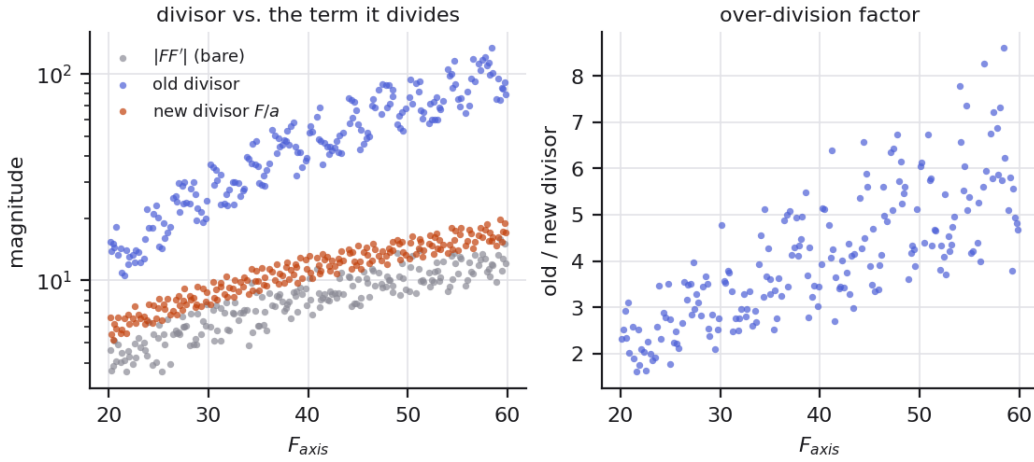


Figure 2: The bare field term is linear in F_{axis} ; the old divisor is quadratic.

Why. With $\text{flux_scale} = F_{axis} a$, we get $F' \sim F_{axis}/\text{flux_scale} = 1/a$, so $|FF'| \sim F_{axis}/a$ — linear. The divisor is quadratic, over-dividing by $1.60 \times - 8.61 \times$ across the domain (median $3.93 \times$). The additive $+1$ is dimensionally meaningless and adds $\sim 12\%$ distortion at low F_{axis} .

4. Change

```
# src/engine/physics.py::_grad_shafranov_residual_from_operator
- scale = (config.State.F_axis / config.Geometry.R0) ** 2 + 1.0
+ scale = config.State.F_axis / config.Geometry.a
```

Pressure is deliberately left out of the divisor: it is 1.52% of the field term, and p_0 shows no correlation with the scale-free error.

Absolute $|R_{GS}|$ rises $\sim 3.9 \times$ at the median for unchanged weights ($2.5 \times 10^{-4} \rightarrow 9.6 \times 10^{-4}$). Nothing measured before the change is comparable to anything after it; the full checkpoint history was re-scored on the common protocol.

5. Confirmation

Does training against the corrected residual produce a better model, or merely a differently-reported one?

The preceding study’s two best configurations were retrained from scratch under the corrected loss. Everything else held: 5×256 MLP, 2,400 epochs, same batch size, seed protocol, and evaluation. Config dicts verified identical field-by-field; **d5b421d** introduces no **src/** changes over **a2b9b73**, so the normaliser is the only source difference.

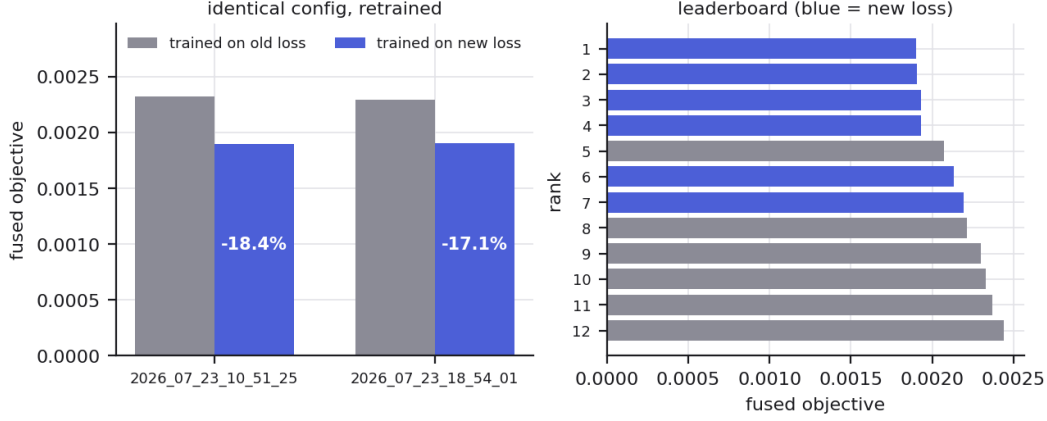


Figure 3: Controlled retrains (left) and the re-scored leaderboard (right).

Table 1: Fused objective, median + 0.3 p_{95} .

trial	old loss	new loss	change
2026-07-23 10:51	2.3278×10^{-3}	1.9005×10^{-3}	-18.4%
2026-07-23 18:54	2.2972×10^{-3}	1.9051×10^{-3}	-17.1%

Both take rank 1 and 2 of all 77 distinct re-scored checkpoints.

Against noise: a 10-trial Optuna search seeded by these two returned a best of 1.9306×10^{-3} — 1.6% *worse* than its own seed. The top four span 1.6%, an indirect bound on single-seed variation (not a repeated-seed measurement). The 17–18% effect sits an order of magnitude above it; the search result sits inside it.

So: the correction is a real gain in the trained model, and the LR/weight-decay/sigma space is exhausted — ~9½ GPU-hours of search failed to beat its starting point.

200 Sobol configurations $\times$ 8,192 area-uniform points, objective median + 0.3 p_{95} of $|R_{GS}|$, seed 42. RTX 3060, JAX with `jax_default_matmul_precision="highest"`; TF32 matmuls inject a $\sim 10^{-2}$ noise floor because the GS operator’s second derivatives amplify boundary-fit harmonic m by $\sim m^2$.